

4.4 How far? – entropy

Students will be assessed on their ability to:

- a demonstrate an understanding that, since endothermic reactions can occur spontaneously at room temperature, enthalpy changes alone do not control whether reactions occur
- b demonstrate an understanding of entropy in terms of the random dispersal of molecules and of energy quanta between molecules
- c demonstrate an understanding that the entropy of a substance increases with temperature, that entropy increases as solid $\rightarrow$ liquid $\rightarrow$ gas and that perfect crystals at zero kelvin have zero entropy
- d demonstrate an understanding that the standard entropy of a substance depends mainly on its physical state but also on its complexity
- e demonstrate an understanding that reactions occur due to chance collisions, and that one possible ordered arrangement, e.g. in a crystalline solid, can be rearranged into many possible disordered arrangements, e.g. in a solution, so the probability of disorder is greater than order
- f interpret the natural direction of change as being in the direction of increasing total entropy (positive entropy change), e.g. gases spread spontaneously through a room
- g carry out experiments and relate the results to disorder and enthalpy changes including:
 - i dissolving a solid, e.g. adding ammonium nitrate crystals to water
 - ii gas evolution, e.g. reacting ethanoic acid with ammonium carbonate
 - iii exothermic reaction producing a solid, e.g. burning magnesium ribbon in air
 - iv endothermic reaction of two solids, e.g. mixing solid barium hydroxide, $\text{Ba}(\text{OH})_2 \cdot 8\text{H}_2\text{O}$ with solid ammonium chloride
- h demonstrate an understanding that the entropy change in any reaction is made up of the entropy change in the system added to the entropy change in the surroundings, summarised by the expression:

$$\Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$$

- i calculate the entropy change in the system for a reaction, ΔS_{system} , given entropy data
- j use the expression $\Delta S_{\text{surroundings}} = -\Delta H/T$ to calculate the entropy change in the surroundings and hence ΔS_{total}
- k demonstrate an understanding that the feasibility of a reaction depends on the balance between ΔS_{system} and $\Delta S_{\text{surroundings}}$, and that at higher temperatures the magnitude of $\Delta S_{\text{surroundings}}$ decreases and its contribution to ΔS_{total} is less. Reactions can occur as long as ΔS_{total} is positive even if one of the other entropy changes is negative
- l demonstrate an understanding of and distinguish between the concepts of thermodynamic stability and kinetic inertness
- m calculate ΔS_{system} and $\Delta S_{\text{surroundings}}$ for the reactions in 4.4g to show that endothermic reactions can occur spontaneously at room temperature
- n define the term *enthalpy of hydration* of an ion and use it and lattice energy to calculate the enthalpy of solution of an ionic compound
- o demonstrate an understanding of the factors that affect the values of enthalpy of hydration and the lattice energy of an ionic compound
- p use entropy and enthalpy of solution values to predict the solubility of ionic compounds.